How-To Library

Informatica

Using a Kerberized Microsoft SQL Server Connection in a Kerberos Environment in Test Data Management

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Abstract

You can use Test Data Management (TDM) to perform tasks with Kerberized Microsoft SQL Server connections in a Kerberos environment. This article describes the steps to create a Kerberized Microsoft SQL Server connection and perform data masking when Informatica service and Test Data Management are installed in a Kerberos environment.

Supported Versions

- Test Data Management 10.2.0 HotFix 2
- Test Data Management 10.2.0 HotFix 1
- Test Data Management 10.2 R1
- Test Data Management 10.2.0

Table of Contents

Overview. 2
Example. 2
Prerequisites. 3
Step 1. Configure Property Files for Microsoft SQL Server. 3
Step 2. Create and Test the Connections. 4
Step 3. Import the Source Metadata into the Project. 8
Step 4. Assign the Masking Rule to a Source Column. 9
Step 5. Create a Data Masking Plan. 10
Step 6. Run the Data Masking Plan to Create the Test Data. 11

Overview

You can install and use Test Data Management in a Kerberos environment.

Before you can create and run plans, you must ensure that the machine on which you create the Test Data Manager Service is configured for Kerberos.

You can use a Microsoft SQL Server database that you create in a Kerberos environment with TDM. A Microsoft SQL Server connection requires additional configuration to ensure that you can create and test the connection successfully in a Kerberos environment.

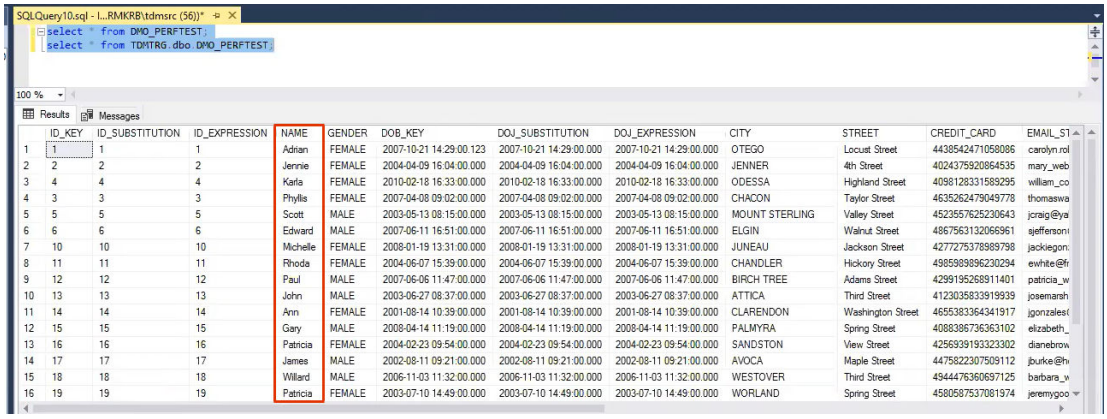
Configure the connection for Kerberos and confirm that you can successfully test the connection before you use it in a plan.

Example

You work for a test team that uses Test Data Management to create and manage test data.

You work in a Kerberos environment, and the Informatica domain and Test Data Management are configured for Kerberos authentication. For a specific project, you need to work with Kerberized Microsoft SQL Server databases. Source metadata that you require is on a Microsoft SQL Server database and you need to create masked data in a target Microsoft SQL Server database. The servers are configured for Kerberos authentication.

The following data represents the sample data used in the example with the column to mask highlighted:



	ID_KEY	ID_SUBSTITUTION	ID_EXPRESSION	NAME	GENDER	DOB_KEY	DOJ_SUBSTITUTION	DOJ_EXPRESSION	CITY	STREET	CREDIT_CARD	EMAIL_ST
1	1	1	1	Adrian	FEMALE	2007-10-21 14:29:00.123	2007-10-21 14:29:00.000	2007-10-21 14:29:00.000	OTEGO	Locust Street	4438542471058086	carolyn.rol
2	2	2	2	Jennie	FEMALE	2004-04-09 16:04:00.000	2004-04-09 16:04:00.000	2004-04-09 16:04:00.000	JENNER	4th Street	4024375920864535	mary_web
3	4	4	4	Karla	FEMALE	2010-02-18 16:33:00.000	2010-02-18 16:33:00.000	2010-02-18 16:33:00.000	ODESSA	Highland Street	4098128331588295	william_co
4	3	3	3	Phyllis	FEMALE	2007-04-08 09:02:00.000	2007-04-08 09:02:00.000	2007-04-08 09:02:00.000	CHACON	Taylor Street	4635262479049778	thomaswa
5	5	5	5	Scott	MALE	2003-05-13 08:15:00.000	2003-05-13 08:15:00.000	2003-05-13 08:15:00.000	MOUNT STERLING	Valley Street	4523557625230643	jcraig@ya
6	6	6	6	Edward	MALE	2007-06-11 16:51:00.000	2007-06-11 16:51:00.000	2007-06-11 16:51:00.000	ELGIN	Walnut Street	4867563132066961	sjefferson
7	10	10	10	Michelle	FEMALE	2008-01-19 13:31:00.000	2008-01-19 13:31:00.000	2008-01-19 13:31:00.000	JUNEAU	Jackson Street	4277275378989798	jackiegon
8	11	11	11	Rhoda	FEMALE	2004-06-07 15:39:00.000	2004-06-07 15:39:00.000	2004-06-07 15:39:00.000	CHANDLER	Hickory Street	4985989896230294	ewhite@fr
9	12	12	12	Paul	MALE	2007-06-06 11:47:00.000	2007-06-06 11:47:00.000	2007-06-06 11:47:00.000	BIRCH TREE	Adams Street	4299195268911401	patricia_w
10	13	13	13	John	MALE	2003-06-27 08:37:00.000	2003-06-27 08:37:00.000	2003-06-27 08:37:00.000	ATTICA	Third Street	4123035833919939	josemarsh
11	14	14	14	Ann	FEMALE	2001-08-14 10:39:00.000	2001-08-14 10:39:00.000	2001-08-14 10:39:00.000	CLARENDON	Washington Street	4655383364341917	jgonzales(
12	15	15	15	Gary	MALE	2008-04-14 11:19:00.000	2008-04-14 11:19:00.000	2008-04-14 11:19:00.000	PALMYRA	Spring Street	4088386736363102	elizabeth_
13	16	16	16	Patricia	FEMALE	2004-02-23 09:54:00.000	2004-02-23 09:54:00.000	2004-02-23 09:54:00.000	SANDSTON	View Street	4256939193323302	dianebrow
14	17	17	17	James	MALE	2002-08-11 09:21:00.000	2002-08-11 09:21:00.000	2002-08-11 09:21:00.000	AVOCA	Maple Street	4475822307509112	jburke@h
15	18	18	18	Willard	MALE	2006-11-03 11:32:00.000	2006-11-03 11:32:00.000	2006-11-03 11:32:00.000	WESTOVER	Third Street	4944476360697125	barbara_w
16	19	19	19	Patricia	FEMALE	2003-07-10 14:49:00.000	2003-07-10 14:49:00.000	2003-07-10 14:49:00.000	WORLAND	Spring Street	4580587537081974	jeremygoo

To use a Kerberized Microsoft SQL Server database with TDM to create test data, perform the following tasks:

1. Configure the Kerberos configuration properties file for Microsoft SQL Server.
2. Create and test connections to the databases in Test Data Manager.
3. Import the source metadata into the required project.
4. Assign the masking rule to the required column.
5. Create and run a data masking plan to create the test data.

Prerequisites

Before you use a Kerberized Microsoft SQL Server with TDM in a Kerberos environment, perform the following tasks:

- Ensure that the Microsoft SQL Server databases are configured for Kerberos authentication.
- Verify that the Test Data Management version supports Kerberos authentication.
- Ensure that the following files are configured on the machine where you create the Test Data Manager Service:
  - `krb5.conf`. Contains configuration information for Kerberos. This file is stored in the following location: `<Informatica installation directory>/TDM/datadirect`.
  - `org.eclipse.virgo.kernel.authentication.config`. Contains authentication properties for the TDM server machine. This file is stored in the following location: `<Informatica installation directory>/TDM/configuration`.
- Ensure that the Kerberos kinit ticket is valid. You cannot run a plan if the ticket is not up to date.
- Create a project and the required masking rules. Add the rules to the project.

Step 1. Configure Property Files for Microsoft SQL Server

You must configure property files on the machine where you create the Test Data Manager Service.

1. Browse to the `<Informatica installation directory>/TDM/datadirect/krb5.conf` file.
   - If the databases and the Informatica domain are in the same Kerberos realm, verify that the file is configured correctly.

- If you work in a Kerberos multi realm environment with cross-realm communication, perform the following steps:
    1. Change the *default_realm* property value to the name of the realm that hosts the Microsoft SQL Server database.
    2. Save the changes and close the file.

2. Browse to the `<Informatica installation directory>/TDM/configuration/org.eclipse.virgo.kernel.authentication.config` file.
3. Add the JDBC driver details in the file and save the changes.

   The following code represents sample information required:

```
JDBC_DRIVER_01 {
          com.sun.security.auth.module.Krb5LoginModule required
           doNotPrompt=<Enter true or false>
           useKeyTab=<Enter true or false>
           keyTab="file:<path to the user keytab file>"
           principal="<The fully qualified user name that must be used as the principal.
For example: <user name>@<realm_name>>"
           storeKey=<Enter true or false>
           debug=<Enter true or false>;
};
```

4. Recycle the Test Data Manager Service after you update the files.

Step 2. Create and Test the Connections

Create connections to the databases that you want to use in a plan.

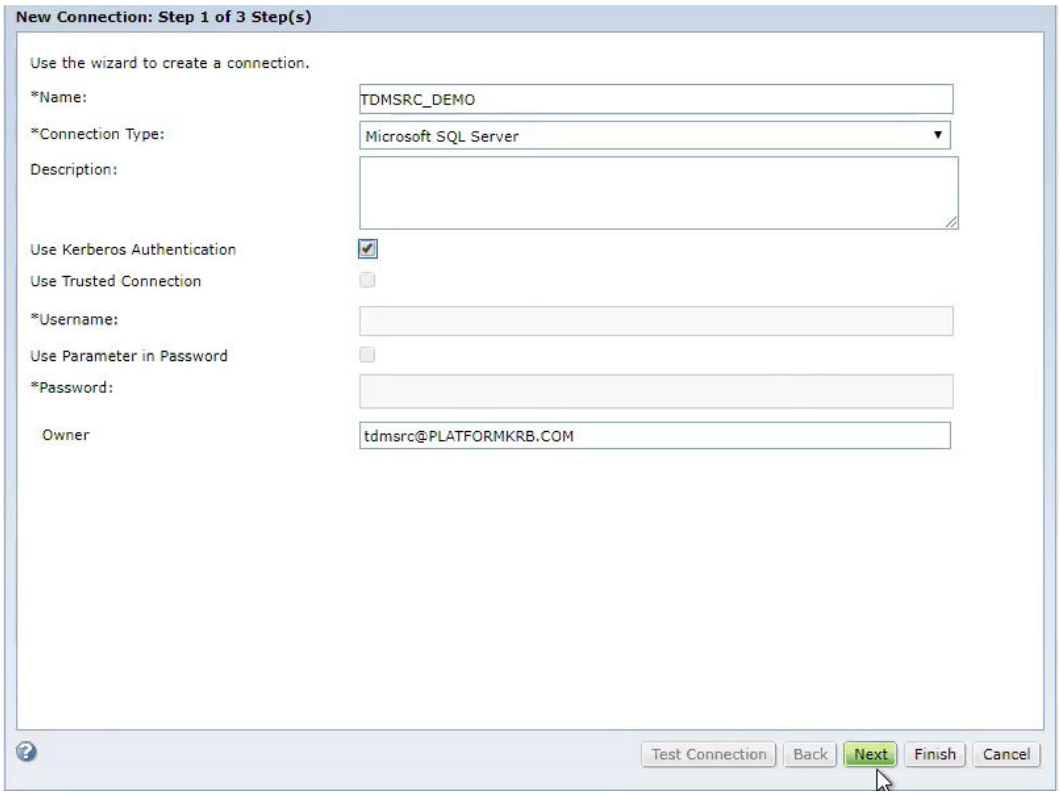
Log in to Test Data Manager as an Administrator to create connections. In a Kerberos authentication environment, you choose the security domain and click Log In.

1. In the **Administrator** | **Connections** view, select **New Connection** from the **Actions** menu.

   A tab opens to display the new connection properties.
2. Select Microsoft SQL Server as the connection type and define the connection name, description, and user information.

   The connection name must begin with an alphabetic character. If you enter a connection name that begins with a numeric character, a workflow that includes the connection might fail.

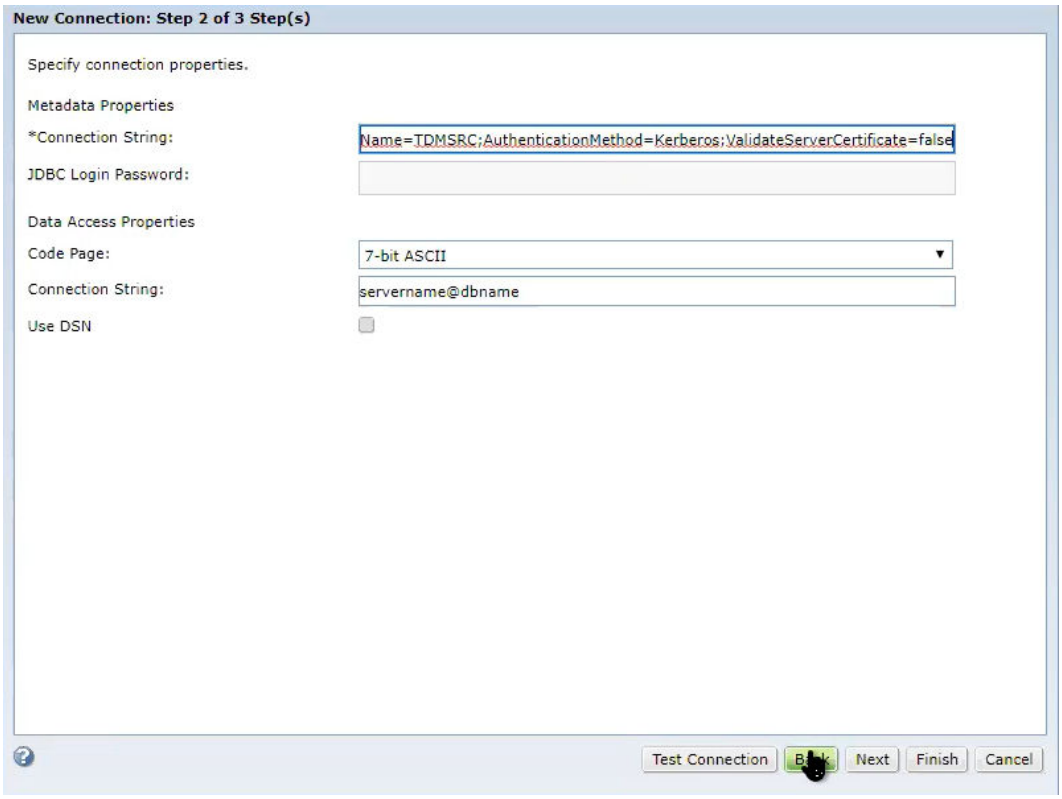
The following image shows the first step in connection creation wizard:



3. Select the **Use Kerberos Authentication** option.

   When you configure a connection to use Kerberos authentication you do not enter a user name or password.
4. Click **Next**.
5. Enter the connection properties.

The following image shows the connection properties page:



a. In the pre populated Connection String property value, enter the appropriate host name and database name values:

```
jdbc:informatica:sqlserver://<hostname>:
1433;SelectMethod=cursor;databaseName=<dbname>
```

b. Append the following properties and values to the connection string:
`AuthenticationMethod=Kerberos;ValidateServerCertificate=false`

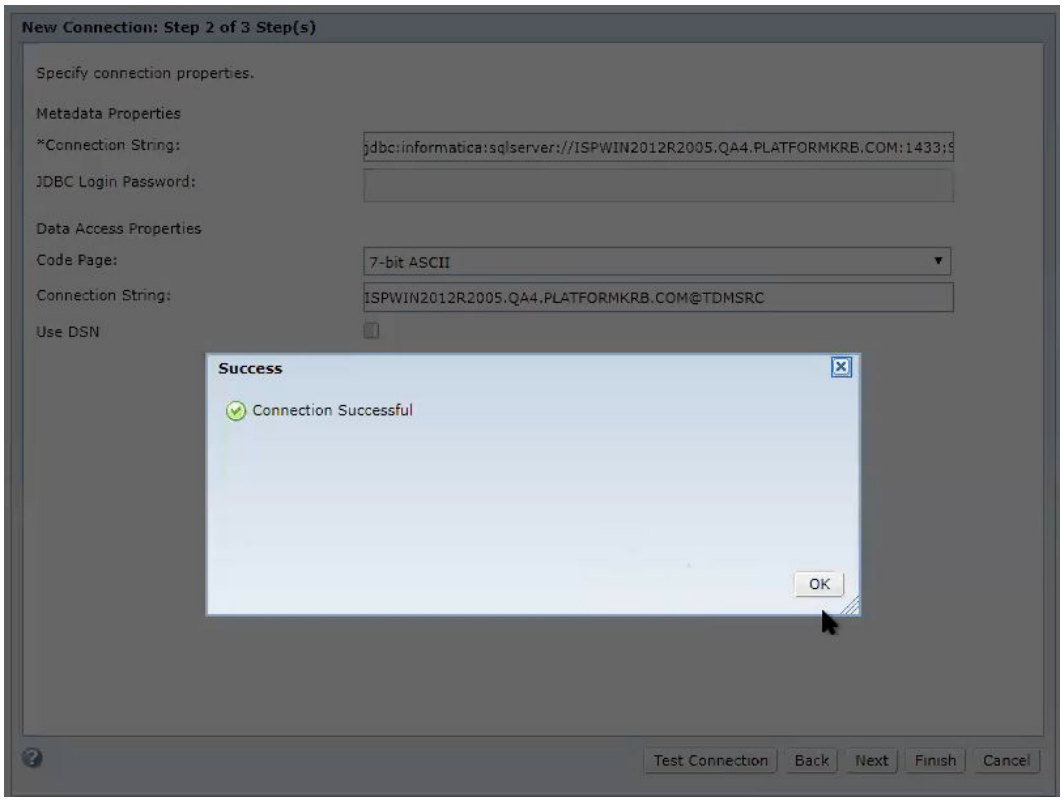
```
jdbc:informatica:sqlserver://<hostname>:
1433;SelectMethod=cursor;databaseName=<dbname>;AuthenticationMethod=Kerberos;Validate
ServerCertificate=false
```

c. Optional. If you enable SSL on the database, append the following property: `EncryptionMethod=SSL`

d. Enter the connection string.

6. Click **Test Connection** to test the connection.

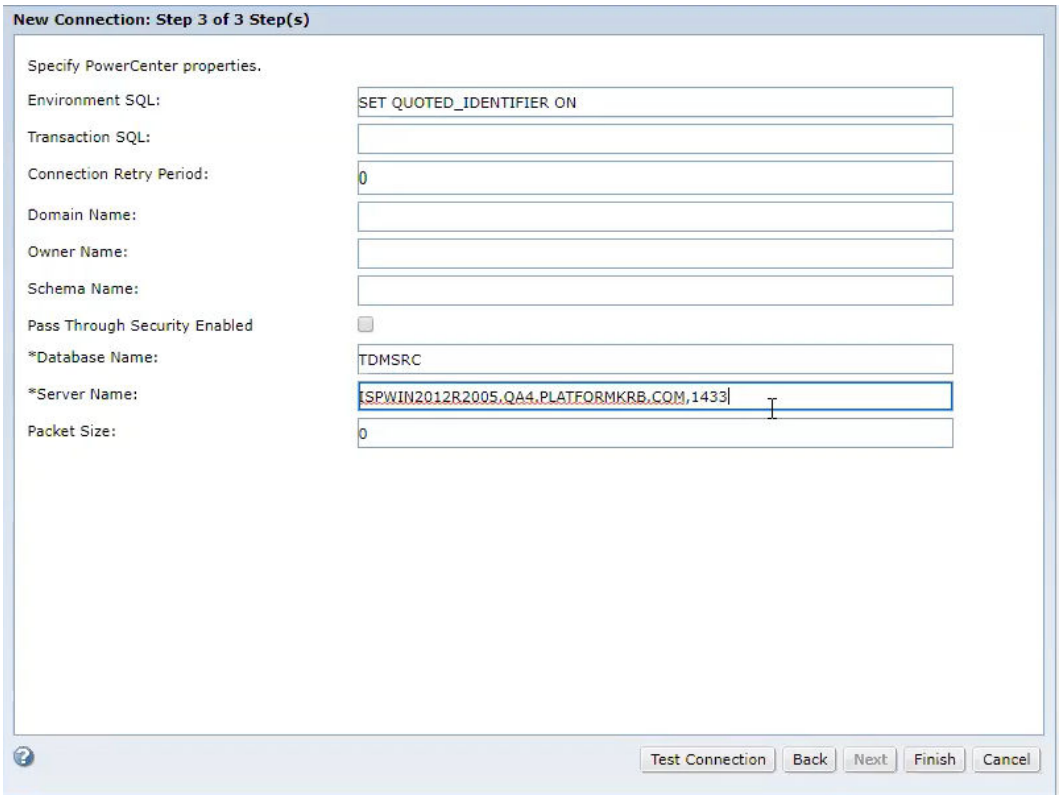
The following image shows a successful test connection message:



Test connection fails if you enter a property value incorrectly or if you do not configure the Kerberos property files.

7. Click **Next**.
8. In the third step of the connection creation wizard, enter the database name and the server name.

The following image shows the third step in connection creation:



Enter the server name in the format: `<hostname>,1433`

9. Click **Finish** to save the connection.

   The connection is visible in the **Administrator | Connections** view.

10. Optional. Select the connection in the **Administrator | Connections** view, and click **Actions** > **Test Connection** to validate the connection.

Repeat the steps to create the target connection.

Note: If you edit the connection properties after you create the connection, the **Save** button is unavailable. After you make the changes, uncheck the **Use Kerberos Authentication** option and select it again. The **Save** button becomes available.

Step 3. Import the Source Metadata into the Project

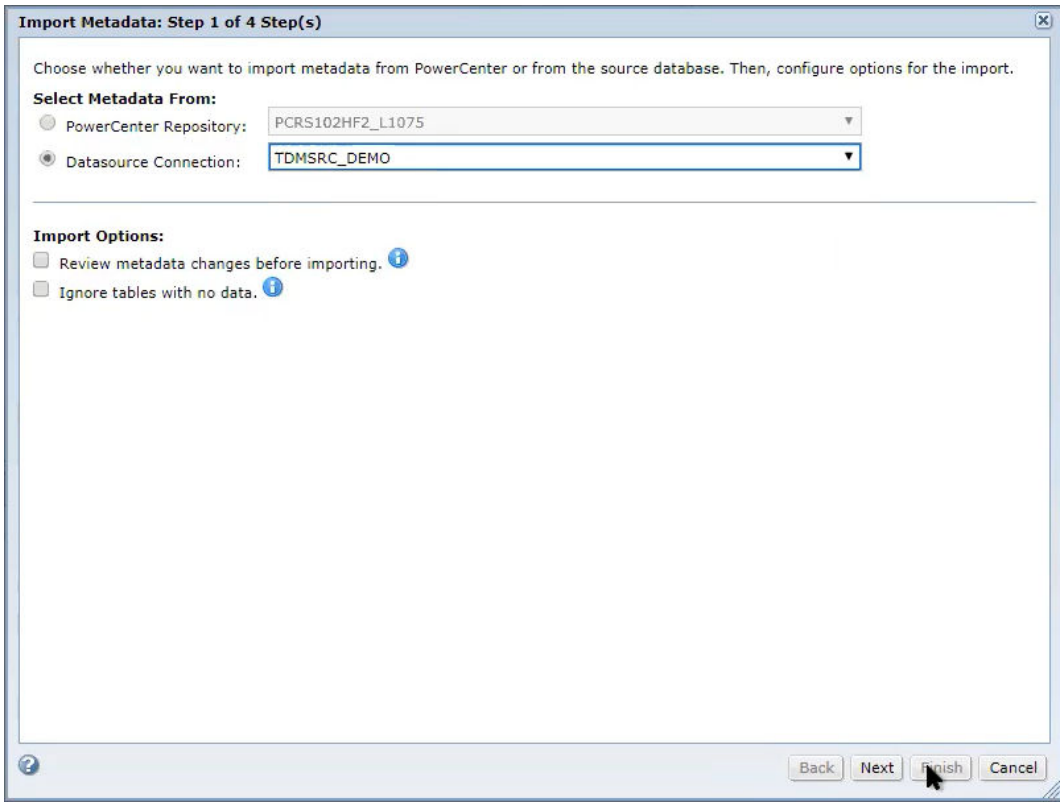
To use the source metadata in a plan, add the metadata to the project.

1. To view a list of projects in Test Data Manager, click **Projects**.
2. Click the required project in the list to open it.
3. Click **Actions** > **Import Metadata**.

   The **Import Metadata** window appears.

4. Choose to import metadata from a data source connection and select the connection from the list.

The following image shows the first step of metadata import:



5. Click **Next**.
6. Select the schema to import. You can filter schemas by schema name.
7. Click **Next**.
8. Select the tables that you want to import. You can filter the tables by data source, table name, or table description.
9. Click **Next**.
10. Choose **Import Now** to import the data source immediately.
11. Click **Finish**.

    The **Import Progress** dialog box appears with the import progress bar.

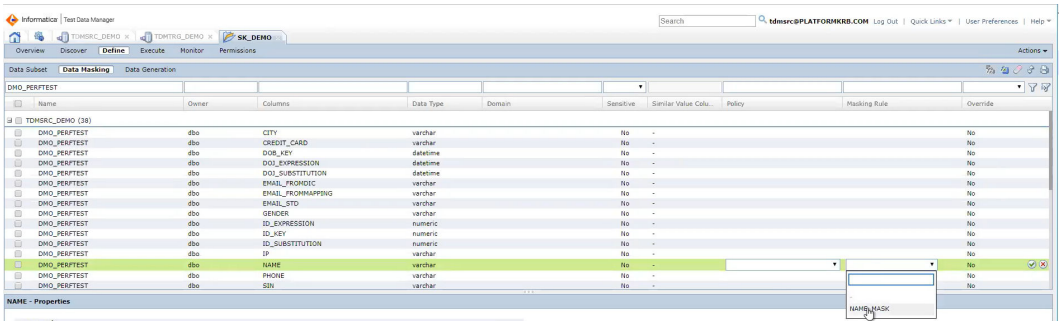
After the job finishes, access the imported metadata through the **Data Sources** details view.

Step 4. Assign the Masking Rule to a Source Column

Assign the masking rules that you created and added to the project to the required columns.

1. In the project, click **Define** | **Data Masking** to access the **Data Masking** view.
2. Select the column that you want to mask.
3. Click inside the **Rule** column to view a list of available rules.
4. Select the masking rule that you want to assign. If you choose a substitution masking rule, you can specify the rule assignment parameters. If you choose a mapplet or advanced rule, you can assign the rule columns to the columns in the project.

The following image shows the Data Masking tab on the Define page:



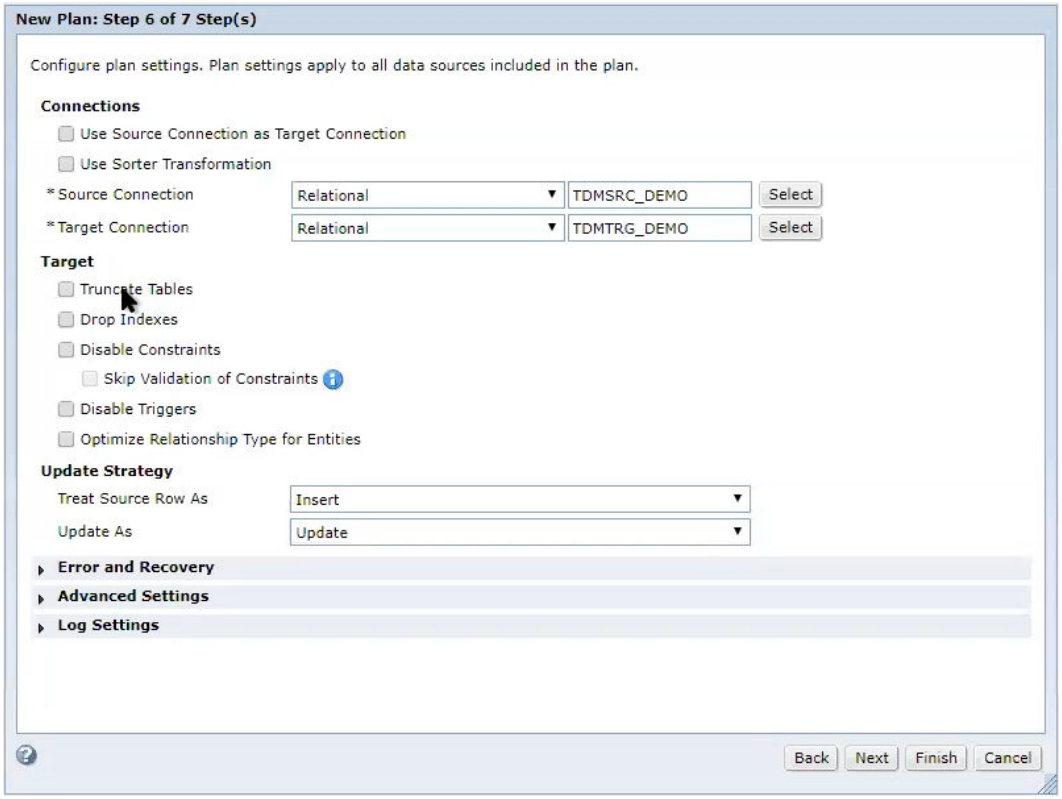
5. Click **Save** for each column that you update.

Step 5. Create a Data Masking Plan

Create a plan and add the required components.

1. Open a project and click **Execute** to view the project plans.
2. Click **Actions** > **New**.
3. In the **New Plan** dialog box, enter a name and optional description for the plan.
4. To add a data masking operation to the plan, click **Add Masking Components**.
5. Select the masking rules to add to the plan and click **Next**.
6. To skip a masking rule, select the check box for the rule and click **Off**.
7. Configure the source and target connections and other properties. Click **Next**.

   The following image shows the **Plan Settings** section of the plan creation wizard:



8. To override plan settings for a data source, select the data source and click **Override Plan Settings** and enter the properties.
9. To override data source settings for a table, select the table and click **Override Data Source Settings** and enter the properties.
10. Click **Finish**.

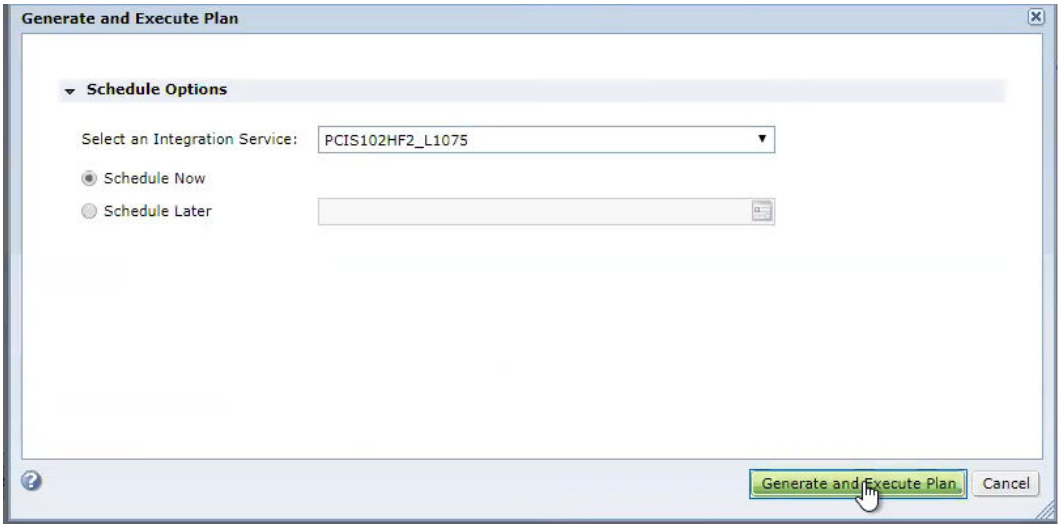
    The plan appears in the project.

Step 6. Run the Data Masking Plan to Create the Test Data

Run the plan to create test data in the target database.

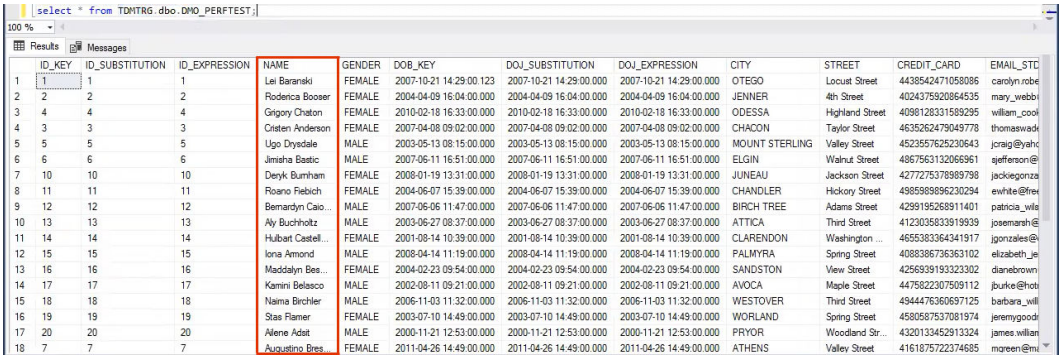
1. To generate and run a workflow in a single step, in the **Plan | Properties** page, click **Actions** > **Generate and Execute**.
2. Choose the required Integration Service.
3. Choose to run the workflow immediately or schedule it for a later time.
4. Click the **Generate and Execute** button to run the workflow.

   The following image shows the **Generate and Execute Plan** dialog box:



5. View the workflow status in the **Workflow Execution** view.
6. View the status of the job on the **Monitor** tab.
7. When the job finishes successfully you can view the masked data in the target database.

   The following image shows the masked sample data:



Note: If you run the workflow with the **Disable Constraints** option selected, and TDM fails to enable the constraints after loading the data, the workflow fails. You must then manually enable the constraints in the database and run the workflow again.

Author

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Acknowledgements

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